Supplementary Material

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1 Implementation Details

All experiments are conducted on a workstation equipped with an Intel Core i7-14700KF CPU, 64GB RAM, and an NVIDIA GPU with 12GB memory. For all LLM-based methods, we adopt GPT-2 with six Transformer layers as the default language model backbone, where all backbone parameters are kept frozen during training to ensure fair comparison and computational efficiency. Unless otherwise specified, the input sequence length is fixed to $L = 512$ across all methods and datasets. All datasets are evaluated under multiple forecasting horizons $H \in 96, 192, 336, 720$ to assess robustness across different prediction scales. The Adam optimizer is employed to minimize the forecasting loss, with the learning rate selected from $1 \times 10^{-3}, 5 \times 10^{-4}, 1 \times 10^{-4}$ based on validation performance. For fair comparison, all methods are trained under identical experimental settings and hyperparameter tuning protocols. Each experiment of **AAS** is repeated three times with different random seeds, and the averaged results are reported to reduce randomness and improve statistical reliability. Additional experimental details are provided in the supplementary materials.

2 Long-term Forecasting

Table 1 provides the complete long-term forecasting results with an additional average row aggregated over all prediction horizons ($H \in 96, 192, 336, 720$). Consistent with the observations in the main paper, AAS maintains stable superiority across all horizons. In particular, the full results reveal that the performance variance across different datasets remains small, further confirming the robustness of the proposed method. We also observed that on the weather dataset, AAS did not achieve absolute optimal performance across all prediction steps. This behavior is mainly attributed to the strong periodicity, high noise level, and dense multivariate correlations inherent in Weather data, where short-term dynamics dominate predictive accuracy and reduce the marginal gain brought by semantic-guided temporal selection. In such scenarios, the selector may receive weaker and less stable attribution signals.

3 Few-shot Forecasting

Table 2 reports the complete few-shot forecasting results under the strict setting of using only 10% of the training data, with evaluation conducted on unchanged validation and test sets. Consistent trends are observed across all datasets and forecasting horizons, where AAS consistently achieves lower MSE and MAE than competing methods. These full results further validate the robustness and stability of AAS under data-scarce conditions. Moreover, the performance margin of AAS becomes increasingly pronounced as the prediction horizon extends, with extreme gains on the ETTh2 and ETTm2 datasets. For instance, on ETTh2, AAS achieves an average reduction of 18% in MSE and 13% in MAE compared with the strongest LLM-based baseline TimeCMA, and reduces MSE and MAE by 13% and 9%, respectively, relative to the non-LLM baseline DLinear.

4 Ablation Study

Table 3 reports the complete ablation results of all model variants across all datasets, extending the partial results presented in the main paper. All ablation variants are evaluated under identical experimental settings as the full model. We analyze the model from the following perspectives: (1) w/o Fusion-based Cross-Attention: The SGS module is replaced by a standard cross-attention mechanism for cross-modal fusion. Temporal features serve as queries, while semantic prompt embeddings function as keys and values. Unlike SGS, this variant does not introduce an explicit temporal selection mechanism, but directly produces fused temporal representations via cross-attention. (2) w/o LER: The SGS module is retained, but the LER on the attribution weights is removed. In this setting, semantic guidance is still applied through cross-attention-based temporal selection, while the attribution distribution is unconstrained. (3) w/o SGS: This variant serves as the backbone baseline, obtained by removing the SGS and the LER from the full model. The supplementary table is included to provide comprehensive quantitative evidence and facilitate reproducibility, while detailed analysis and discussions of the ablation results are referred to the main text.

5 Hyperparameter Sensitivity

Look-back Window. Table 4 reports the sensitivity of different look-back window sizes across all four ETT datasets. Overall, consistent trends are observed across datasets. Moderate window lengths achieve the best trade-off between contextual coverage and noise accumulation, while excessively long windows tend to yield marginal or even negative gains, especially under long-horizon forecasting settings. This suggests that although longer histories provide additional temporal context, overly large windows may introduce redundant or weakly relevant patterns that dilute discriminative signals and hinder effective temporal selection.

The ETT datasets are chosen for systematic evaluation due to their balanced scale and diverse temporal dynamics, which allow reliable assessment of long-window behavior without incurring prohibitive training costs or insufficient sequence coverage. Across datasets, the optimal window size remains relatively stable, indicating that the proposed selection mechanism exhibits robust behavior with respect to input length variation. All experimental settings and evaluation protocols follow those of the main experiments, and the table provides complementary quantitative evidence supporting the trend analysis discussed in the main text.

Entropy Control Parameter. Table 5 summarizes the sensitivity of the entropy control parameter α across all datasets. Across different datasets and forecasting horizons, a consistent performance pattern is observed: moderate values of α generally achieve superior or near-optimal performance, while overly small or excessively large values tend to degrade accuracy.

When α is small, the sharpened prior remains close to the original attribution distribution, leading to a weaker regularization effect and limited enhancement of temporal selectivity. In contrast, large α values enforce overly peaked priors, which may overly constrain the selector and reduce its flexibility in modeling diverse temporal dependencies. Importantly, the performance variation across datasets remains relatively limited within a reasonable range of α, indicating that the proposed regularization strategy is robust and does not require dataset-specific tuning. These results further support the role of low-entropy regularization as a stable inductive bias for promoting sparse yet controllable temporal selection.

Categories		LLM-based								Transformer-based				CNN-based		Linear-based	
Models		AAS		TimeCMA		CALF		Time-LLM		iTransformer		PatchTST		TimesNet		Dlinear	
Metrics		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	**0.364**	**0.392**	0.399	0.415	0.387	0.414	0.389	0.410	0.396	0.421	0.392	0.416	0.448	0.460	0.367	0.393
	192	**0.397**	**0.412**	0.418	0.437	0.411	0.429	0.419	0.439	0.426	0.442	0.454	0.461	0.495	0.491	0.402	0.419
	336	**0.426**	**0.435**	0.467	0.452	0.430	0.442	0.453	0.459	0.461	0.466	0.485	0.479	0.536	0.521	0.431	0.443
	720	**0.460**	0.480	0.463	**0.466**	0.476	0.484	0.445	0.465	0.681	0.598	0.672	0.576	0.578	0.539	0.475	0.497
	Avg	**0.412**	**0.430**	0.437	0.443	0.426	0.442	0.427	0.443	0.491	0.482	0.501	0.483	0.514	0.503	0.419	0.438
ETTh2	96	**0.273**	**0.341**	0.375	0.408	0.297	0.353	0.303	0.356	0.302	0.359	0.315	0.371	0.361	0.408	0.300	0.365
	192	**0.328**	**0.376**	0.370	0.412	0.361	0.392	**0.357**	**0.388**	0.373	0.404	0.391	0.417	0.409	0.441	0.396	0.427
	336	**0.349**	**0.393**	0.437	0.463	0.390	0.417	0.375	0.412	0.418	0.435	0.443	0.449	0.395	0.438	0.498	0.491
	720	**0.381**	**0.426**	0.497	0.496	0.411	0.440	0.416	0.448	0.447	0.466	0.506	0.497	0.485	0.490	0.806	0.636
	Avg	**0.333**	**0.384**	0.420	0.445	0.365	0.400	0.363	0.401	0.385	0.416	0.414	0.434	0.413	0.444	0.500	0.480
ETTm1	96	0.303	**0.348**	0.324	0.365	**0.300**	0.351	0.336	0.371	0.310	0.364	0.305	0.354	0.349	0.381	0.305	0.349
	192	**0.336**	**0.368**	0.373	0.393	0.337	0.376	0.361	0.386	0.348	0.387	0.359	0.394	0.480	0.449	0.337	0.368
	336	**0.366**	0.386	0.407	0.415	0.370	0.397	0.388	0.401	0.378	0.404	0.379	0.405	0.410	0.426	0.366	**0.385**
	720	**0.415**	0.423	0.468	0.448	0.424	0.426	0.433	0.428	0.440	0.441	0.451	0.449	0.468	0.462	0.420	**0.417**
	Avg	**0.355**	0.381	0.393	0.405	0.357	0.387	0.380	0.397	0.369	0.399	0.374	0.401	0.427	0.430	0.357	**0.380**
ETTm2	96	**0.162**	**0.253**	0.190	0.275	0.171	0.262	0.187	0.275	0.180	0.272	0.182	0.272	0.194	0.281	0.166	0.261
	192	**0.219**	**0.295**	0.261	0.325	0.227	0.298	0.236	0.309	0.242	0.313	0.236	0.309	0.260	0.324	0.224	0.303
	336	**0.275**	**0.331**	0.317	0.360	0.288	0.335	0.283	0.341	0.294	0.347	0.293	0.346	0.319	0.369	0.296	0.359
	720	**0.367**	**0.387**	0.397	0.411	0.368	0.387	0.368	0.390	0.373	0.395	0.396	0.408	0.421	0.423	0.410	0.432
	Avg	**0.256**	**0.316**	0.291	0.343	0.263	0.320	0.269	0.329	0.272	0.332	0.277	0.334	0.299	0.349	0.274	0.339
Weather	96	0.168	0.223	0.168	0.225	0.158	0.209	0.157	**0.209**	0.169	0.220	**0.155**	0.211	0.164	0.220	0.171	0.232
	192	0.202	**0.234**	0.203	0.256	0.204	0.249	**0.198**	0.247	0.212	0.257	0.209	0.259	0.221	0.270	0.215	0.274
	336	**0.245**	**0.282**	0.247	0.286	0.250	0.285	0.250	0.287	0.268	0.295	0.248	0.286	0.281	0.313	0.259	0.292
	720	0.321	0.335	**0.311**	**0.334**	0.332	0.342	0.317	0.334	0.334	0.341	0.318	0.336	0.344	0.356	0.320	0.359
	Avg	0.234	**0.268**	0.232	0.275	0.236	0.271	**0.231**	0.269	0.246	0.278	0.233	0.273	0.253	0.290	0.241	0.289
Exchange	96	**0.086**	**0.205**	0.263	0.381	0.099	0.227	0.107	0.235	0.107	0.233	0.216	0.248	0.247	0.369	0.117	0.259
	192	**0.168**	**0.298**	0.629	0.603	0.191	0.315	0.252	0.362	0.208	0.327	0.209	1.274	0.375	0.456	0.234	0.368
	336	**0.351**	**0.432**	0.802	0.679	0.423	0.481	0.462	0.504	0.419	0.480	0.378	0.453	0.622	0.601	0.627	0.620
	720	**0.912**	**0.716**	2.328	1.211	1.129	0.808	1.194	0.823	1.020	0.765	1.274	0.834	1.515	0.928	1.170	0.827
	Avg	**0.379**	**0.366**	1.006	0.719	0.460	0.458	0.504	0.481	0.439	0.451	0.519	0.702	0.690	0.589	0.537	0.519

Table 1: Long-term forecasting results of AAS and baseline methods. A look-back window of 512 is used with prediction horizons $H \in \{96, 192, 336, 720\}$. Performance is measured by MSE and MAE, with best values in bold

Categories		LLM-based								Transformer-based				CNN-based		Linear-based	
Models		AAS		TimeCMA		CALF		Time-LLM		iTransformer		PatchTST		TimesNet		Dlinear	
Metrics		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	**0.437**	**0.441**	0.699	0.561	0.545	0.516	0.545	0.496	0.591	0.520	0.546	0.506	0.869	0.636	0.451	0.453
	192	**0.481**	0.473	0.761	0.569	0.596	0.542	0.691	0.569	0.661	0.546	0.629	0.544	0.883	0.652	0.481	**0.469**
	336	0.549	0.524	0.758	0.586	0.763	0.594	1.029	0.716	0.731	0.636	0.724	0.599	0.866	0.640	**0.505**	**0.492**
	720	**0.659**	**0.573**	0.751	0.613	0.904	0.668	0.957	0.663	0.849	0.650	0.798	0.646	0.883	0.663	0.692	0.599
	Avg	**0.531**	**0.502**	0.742	0.582	0.702	0.580	0.806	0.611	0.708	0.588	0.674	0.574	0.875	0.648	0.532	0.503
ETTh2	96	**0.273**	**0.346**	0.383	0.426	0.340	0.397	0.373	0.412	0.358	0.401	0.384	0.425	0.464	0.476	0.335	0.384
	192	**0.328**	**0.386**	0.441	0.463	0.515	0.490	0.466	0.480	0.420	0.438	0.460	0.456	0.481	0.489	0.421	0.441
	336	**0.371**	**0.413**	0.447	0.463	0.496	0.489	0.572	0.521	0.434	0.454	0.493	0.485	0.463	0.474	0.438	0.459
	720	**0.464**	**0.470**	0.484	0.489	0.485	0.482	0.481	0.478	0.491	0.495	0.463	0.473	0.503	0.502	0.466	0.496
	Avg	**0.359**	**0.403**	0.439	0.460	0.459	0.464	0.473	0.473	0.426	0.447	0.450	0.460	0.478	0.485	0.415	0.445
ETTm1	96	**0.349**	**0.378**	0.527	0.484	0.397	0.414	0.418	0.418	0.447	0.447	0.501	0.463	0.717	0.573	0.362	0.396
	192	**0.371**	**0.391**	0.660	0.551	0.475	0.456	0.486	0.451	0.515	0.486	0.547	0.478	0.783	0.605	0.387	0.411
	336	**0.403**	**0.410**	0.768	0.592	0.522	0.488	0.520	0.469	0.584	0.519	0.579	0.496	0.835	0.599	0.418	0.430
	720	0.512	0.479	0.849	0.633	0.597	0.527	0.675	0.550	0.660	0.558	0.957	0.643	0.867	0.619	**0.490**	**0.476**
	Avg	**0.408**	**0.414**	0.701	0.565	0.498	0.471	0.525	0.472	0.552	0.503	0.646	0.520	0.801	0.599	0.414	0.428
ETTm2	96	**0.178**	**0.262**	0.256	0.327	0.190	0.268	0.221	0.298	0.203	0.289	0.218	0.302	0.282	0.346	0.207	0.294
	192	**0.235**	**0.299**	0.293	0.349	0.258	0.311	0.267	0.329	0.262	0.329	0.283	0.342	0.323	0.370	0.265	0.331
	336	**0.290**	**0.336**	0.343	0.389	0.320	0.349	0.329	0.366	0.309	0.358	0.342	0.376	0.382	0.406	0.318	0.367
	720	**0.384**	**0.393**	0.483	0.469	0.443	0.422	0.438	0.429	0.411	0.418	0.469	0.447	0.500	0.473	0.400	0.416
	Avg	**0.271**	**0.322**	0.344	0.384	0.303	0.338	0.314	0.356	0.296	0.349	0.328	0.367	0.372	0.399	0.298	0.352

Table 2: Few-shot forecasting results. All results are averaged over four prediction horizons $H \in \{96, 192, 336, 720\}$ with a look-back window of 512. Performance is evaluated by MSE and MAE, where the best results are in bold.

Variant		+SGS&LER		+SGS		Fusion-based		Backbone	
Metrics		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	**0.364**	**0.392**	0.373	0.400	0.395	0.416	0.378	0.3985
	192	**0.397**	**0.412**	0.404	0.420	0.451	0.457	0.411	0.419
	336	**0.426**	**0.435**	0.429	0.437	0.470	0.466	0.438	0.441
	720	**0.460**	**0.480**	0.468	0.483	0.543	0.590	0.483	0.507
	Avg	**0.412**	**0.430**	0.418	0.435	0.465	0.482	0.444	0.441
ETTh2	96	**0.273**	**0.341**	0.280	0.348	0.294	0.350	0.284	0.350
	192	**0.328**	**0.376**	0.331	0.382	0.349	0.391	0.337	0.387
	336	**0.349**	**0.393**	0.354	0.400	0.369	0.408	0.362	0.399
	720	**0.381**	**0.426**	0.391	0.432	0.414	0.444	0.398	0.432
	Avg	**0.333**	**0.384**	0.339	0.3905	0.356	0.398	0.345	0.392
ETTm1	96	**0.303**	**0.348**	0.315	0.358	0.320	0.361	0.33	0.369
	192	**0.336**	**0.368**	0.346	0.376	0.342	0.382	0.363	0.383
	336	**0.366**	**0.386**	0.388	0.400	0.393	0.405	0.391	0.407
	720	**0.415**	**0.423**	0.423	0.412	0.459	0.443	0.435	0.432
	Avg	**0.355**	**0.381**	0.368	0.3865	0.378	0.397	0.380	0.398
ETTm2	96	**0.162**	**0.253**	0.170	0.262	0.222	0.303	0.172	0.264
	192	**0.219**	**0.295**	0.224	0.287	0.251	0.319	0.243	0.301
	336	**0.275**	**0.331**	0.281	0.348	0.303	0.352	0.341	0.395
	720	**0.367**	**0.387**	0.391	0.397	0.470	0.448	0.417	0.422
	Avg	**0.256**	**0.316**	0.266	0.323	0.311	0.355	0.293	0.346
Weather	96	**0.168**	**0.223**	0.189	0.247	0.203	0.264	0.195	0.262
	192	**0.202**	**0.234**	0.218	0.271	0.231	0.278	0.221	0.268
	336	**0.245**	**0.282**	0.263	0.298	0.279	0.311	0.267	0.298
	720	**0.321**	**0.335**	0.337	0.353	0.351	0.365	0.355	0.371
	Avg	**0.234**	**0.268**	0.251	0.292	0.266	0.304	0.260	0.299
Exchange	96	**0.086**	**0.205**	0.094	0.218	0.114	0.243	0.114	0.242
	192	**0.168**	**0.298**	0.182	0.313	0.261	0.363	0.196	0.317
	336	**0.351**	**0.432**	0.363	0.425	0.444	0.491	0.406	0.486
	720	**0.912**	**0.716**	0.946	0.73	0.997	0.753	0.998	0.782
	Avg	**0.379**	**0.366**	0.396	0.421	0.454	0.462	0.429	0.457

Table 3: Ablation study of different key modules on all datasets over four prediction horizons (96, 192, 336, 720).

Categories		Forecasting performance of AAS with varying look-back window									
Parameters		96		128		336		512		720	
Metrics		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	0.371	0.398	0.368	0.401	0.369	0.400	**0.364**	**0.392**	0.368	0.397
	192	0.405	0.415	0.401	0.413	0.399	0.416	**0.397**	**0.412**	0.400	0.418
	336	0.430	0.440	0.429	0.439	0.428	0.442	**0.426**	**0.435**	0.429	0.438
	720	0.464	0.483	0.462	0.484	0.463	0.482	**0.460**	**0.480**	0.463	0.485
	Avg	0.418	0.434	0.415	0.434	0.415	0.435	**0.412**	**0.430**	0.415	0.434
ETTh2	96	0.276	0.348	0.276	0.343	0.274	0.343	**0.273**	**0.341**	0.276	0.344
	192	0.331	0.385	**0.327**	0.382	0.330	0.379	0.328	**0.376**	0.331	0.380
	336	0.354	0.396	0.351	0.397	**0.350**	**0.393**	0.350	0.393	0.353	0.400
	720	0.385	0.432	0.388	0.431	0.385	0.431	**0.381**	**0.426**	0.384	0.428
	Avg	0.336	0.390	0.335	0.388	0.335	0.386	**0.333**	**0.384**	0.336	0.388
ETTm1	96	0.307	0.351	0.310	0.351	0.306	0.356	**0.303**	**0.348**	0.306	0.351
	192	0.344	0.376	0.343	0.370	0.341	0.370	**0.336**	**0.368**	0.339	0.372
	336	0.368	0.390	**0.363**	0.401	0.372	0.385	0.366	**0.386**	0.370	0.392
	720	0.419	0.425	0.421	**0.420**	0.418	0.427	**0.415**	0.423	0.422	0.430
	Avg	0.360	0.386	0.359	0.385	0.359	0.384	**0.355**	**0.381**	0.359	0.386
ETTm2	96	0.167	0.258	0.164	0.254	0.166	0.253	**0.162**	**0.253**	0.165	0.258
	192	0.224	0.298	0.223	0.300	0.220	0.299	**0.219**	**0.295**	0.226	0.298
	336	0.278	0.338	0.277	0.334	0.277	0.334	0.275	**0.331**	**0.274**	0.337
	720	0.371	0.391	0.369	0.395	0.369	0.388	**0.367**	**0.387**	0.375	0.396
	Avg	0.260	0.321	0.258	0.321	0.258	0.319	**0.256**	**0.316**	0.260	0.322

Table 4: Forecasting performance of AAS with varying look-back windows.

Categories		Forecasting performance of AAS with varying entropy control parametes							
Parameters		1		2		4		8	
Metrics		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	0.369	0.401	0.368	0.394	**0.364**	**0.392**	0.365	0.395
	192	0.399	0.421	0.400	**0.412**	**0.397**	**0.412**	0.400	0.421
	336	0.427	0.442	0.427	0.437	**0.426**	**0.435**	0.428	0.437
	720	0.463	0.488	0.463	**0.479**	**0.460**	0.480	0.464	0.481
	Avg	0.414	0.438	0.415	0.431	**0.412**	**0.430**	0.414	0.434
ETTh2	96	0.280	0.345	0.279	0.341	**0.273**	**0.341**	0.277	0.350
	192	0.329	0.381	0.335	0.384	**0.328**	**0.376**	0.330	0.379
	336	0.357	0.401	0.353	0.393	**0.349**	**0.393**	0.351	0.401
	720	0.387	0.432	0.383	0.430	**0.381**	**0.426**	0.381	0.428
	Avg	0.338	0.390	0.338	0.387	**0.333**	**0.384**	0.335	0.390
ETTm1	96	0.309	0.353	0.306	0.355	**0.303**	**0.348**	0.308	0.352
	192	0.340	0.375	0.341	0.344	**0.336**	**0.368**	0.338	0.372
	336	0.369	0.391	0.369	0.393	**0.366**	0.386	0.368	**0.385**
	720	0.420	0.427	0.417	0.417	0.415	**0.423**	**0.414**	0.429
	Avg	0.359	0.386	0.358	0.377	**0.355**	**0.381**	0.357	0.385
ETTm2	96	0.169	0.261	0.163	0.255	**0.162**	**0.253**	0.169	0.256
	192	0.225	0.304	0.223	0.295	**0.219**	**0.295**	0.225	0.303
	336	**0.271**	0.336	0.280	0.332	0.275	**0.331**	0.282	0.338
	720	0.369	0.393	**0.365**	0.391	0.367	**0.387**	0.371	0.392
	Avg	0.259	0.324	0.258	0.318	**0.256**	**0.316**	0.262	0.322
Weather	96	0.173	0.228	0.170	0.225	**0.168**	**0.223**	0.172	0.230
	192	0.208	0.239	0.205	0.204	**0.202**	**0.234**	0.209	0.242
	336	0.246	0.286	0.247	0.288	**0.245**	**0.282**	0.248	0.285
	720	0.328	0.340	0.322	0.322	**0.321**	**0.335**	0.323	0.343
	Avg	0.239	0.273	0.236	0.260	**0.234**	**0.268**	0.238	0.275
Exchange	96	0.092	0.214	0.091	0.205	**0.086**	**0.205**	0.091	0.208
	192	0.176	0.302	0.174	0.305	**0.168**	**0.298**	0.171	0.302
	336	0.356	**0.431**	0.357	0.437	**0.351**	0.432	0.355	0.432
	720	0.914	0.719	0.913	0.718	**0.912**	**0.716**	0.916	0.719
	Avg	0.384	0.417	0.384	0.416	**0.379**	**0.413**	0.383	0.415

Table 5: Forecasting performance of AAS with varying entropy control parametes.